

FROM LIBRARY TO CO-PILOT

Building an Intelligent Infrastructure Assistant with Glass Box AI

Phase 9: How 3 weeks of systematic engineering transformed a static documentation system into an intelligent agent with complete visual transparency



98.7% ACCURACY



GLASS BOX AI



41-238× FASTER

THE FOUNDATION: PHASE 8 COMPLETE

WHERE WE STARTED

Phase 8 delivered 90% retrieval accuracy - elite-tier search performance. Production RAG system: 611 chunks, 85 documents, ~25ms latency. Users could ask "What is Nomad?" and get authoritative ADR-based answers.

WHAT WORKED

- ✓ 90% Retrieval Accuracy
- ✓ Fast Document Search
- ✓ Professional Streamlit UI

THE LIMITATION

- ⚠ Still Just a Library
- ⚠ Couldn't See Live Infra
- ⚠ Black Box Reasoning

The Challenge: Transform from passive knowledge base to active co-pilot

Phase 8 proved retrieval works. Phase 9: Make it intelligent.

THE VISION: FROM LIBRARY TO CO-PILOT

THE GAP

Phase 8 RAG knew what SHOULD be running (from documentation). But it couldn't see what IS running (live infrastructure state).

Need: Actual current state + comparison + diagnosis

LIVE SYSTEM ACCESS

MCP tools for Prometheus, Nomad, and Consul. Query real infrastructure in real-time.

INTENT UNDERSTANDING

Classify user queries (Live, Docs, or Diagnostic) and route intelligently.

VISUAL TRANSPARENCY

Glass Box AI: Watch AI think with step-by-step reasoning and execution timelines.

From "What should be" to "What is + Why + How to fix"

THE THREE-WEEK STRATEGY

Systematic week-by-week capability stacking with measurable milestones.

WEEK 1: MCP TOOLS

Foundation building with 5 operational tools and sub-second responses.

WEEK 2: AGENTIC

Intent classification (98.2%) and multi-step workflows (100% success).

WEEK 3: GLASS BOX

Visual thought traces, timelines, and performance dashboards.

FINAL ACHIEVEMENT

98.7% Accuracy | 190 Passing Tests | Complete Explainability

Building on success: No big-bang integration.

WEEK 1: MCP TOOLS FOUNDATION

THE GOAL

Integrate live infrastructure APIs through Model Context Protocol (MCP).
Security-first: read-only, rate-limited, input-validated, timeout-protected.

`prometheus_query`

Query metrics
Response time, CPU, memory
"What's Heimdall's latency?"

`nomad_job_status`

Orchestration state
Job health, allocations
"Is Traefik running?"

`consul_health`

Service discovery
Health check status
"Are services healthy?"

`check_service_health`

Diagnostic aggregation
Multi-source health view
"Show Heimdall health"

`get_job_health_summary`

Cluster overview
Nomad-wide status
"What's in the cluster?"



ADR THRESHOLD INJECTOR

Metrics aren't enough - users need context. Auto-inject documented thresholds into responses:
"Heimdall: 5000ms (EXCEEDS 2000ms threshold from ADR-007)"
Self-documenting diagnostics!

WEEK 1 RESULTS

- ✓ 179/179 tests passing (100%)
- ✓ Sub-second response times
- ✓ 5 MCP tools operational
- ✓ Safety guardrails working

179/179 tests passing (100%) | Sub-second response times

WEEK 2: AGENTIC REASONING

THE CHALLENGE

System needs to understand intent and orchestrate the right workflow: "Is Traefik running?" vs "Why is Heimdall slow?"



INTENT: 98.2%

Classification into Live, Docs,
Diagnosis with <1ms latency.



CROSS-REF: 100%

Correlation of live metrics with ADR
thresholds automatically.



WORKFLOW: 100%

Successful orchestration of multi-
step diagnostic reasoning.

BREAKTHROUGH: 41-238× FASTER THAN TARGET

Response time: 21-121ms actual vs 5000ms engineering goal.

Intent: 98.2% | Cross-ref: 100% | Reasoning:
100%

WEEK 3: GLASS BOX AI ★

SOLVING THE BLACK BOX

Traditional agents hide their process. Glass Box AI exposes every step, tool call, and duration to the user for absolute trust.



THOUGHT TRACE

Step-by-step reasoning with duration and JSON details.



TIMELINE

Gantt chart visualization of the execution flow.



PERFORMANCE

Dashboard showing bottleneck identification and speedup.

THE BREAKTHROUGH

Glass Box revealed 99.9% of time is in MCP API calls, not AI logic. Directed optimization precisely!

Let's see Glass Box AI in
action...

GLASS BOX AI: THOUGHT TRACE



AG

1

Agentic

fication

)

ive

ta)

hybrid

nce (ADR

elligent

ting step

sk in real-

decisions

trics for

AI in action!

able

ble:

apps/rag_

...

Agentic RAG Chat - Glass Box AI

Watch the AI reason step-by-step with visual thought traces

🔴 Is Traefik running?

🔴 Job: traefik Status: running Type: service Priority: 50 TaskGroup: traefik (Desired: 1)

OK NORMAL: Job status is 'running' per ADR-001

🔴 Thought Trace

📊 Timeline

⚡ Performance

🧠 Thought Trace - How the AI Reasoned

Intent	Total Time	Steps Executed	Success Rate
LIVE_STATUS	1010ms	3	100%
📈 90.0% confident			

Step-by-Step Reasoning

🔴 Step 1: Classify user intent

🕒 0.1ms

View Details

Inputs:

{

"query" : "Is Traefik running?"

}

Outputs:

{

"intent" : "LIVE_STATUS"

"confidence" : 0.5

"reasoning" : "Matched live_status patterns"

"recommended_tools" : [...]

"matched_patterns" : [...]

}

🔍 Step 2: Get traefik job status

🕒 1009.8ms

Ask about your homelab...

OVERALL METRICS

Intent: LIVE_STATUS
Time: 1010ms
Success: 100%

BOTTLENECK FOUND

Step 2: 🛠 nomad_job (1009.8ms)
MCP tool call takes 99.9% of time.

FULL TRANSPARENCY

Expandable JSON details for every input and output step.

Query: "Is Traefik running?" - Watch the AI think

GLASS BOX AI: TIMELINE VISUALIZATION

INTENT: 0.1ms

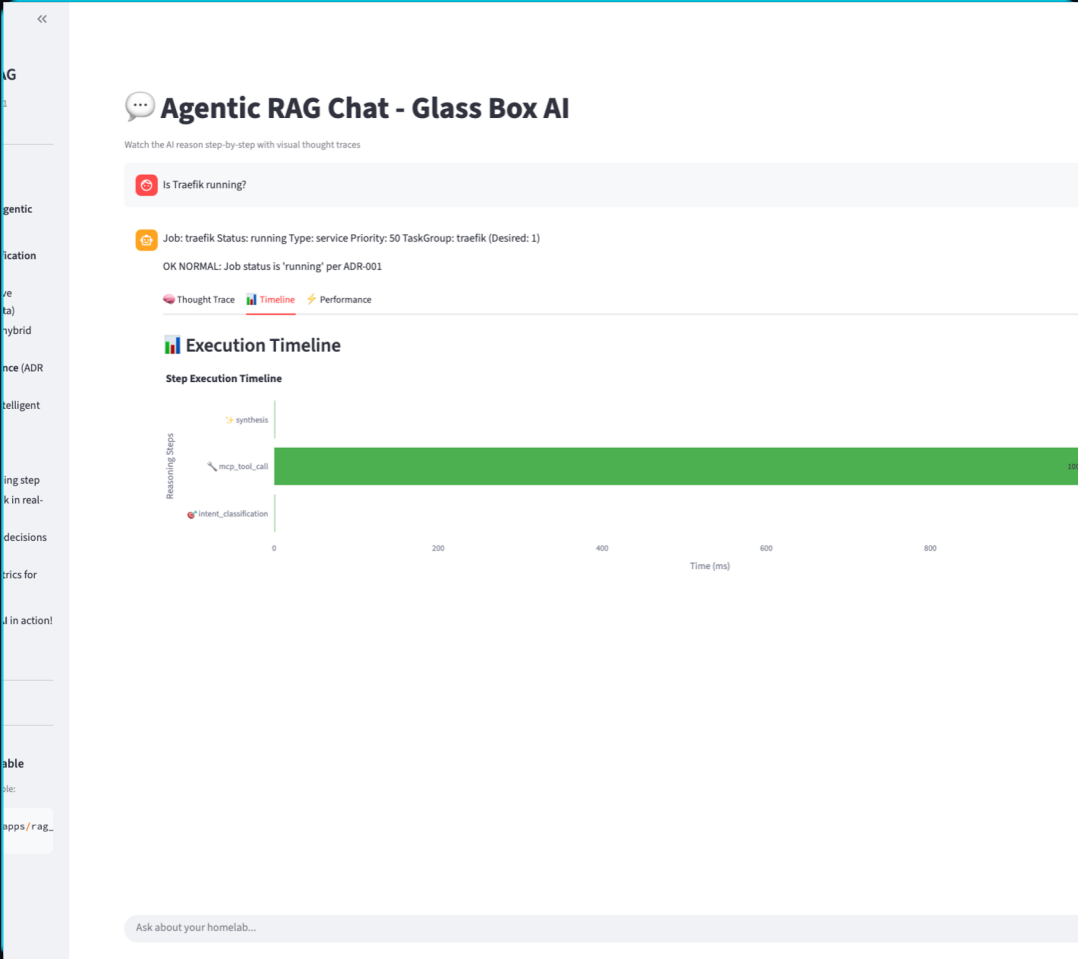
0.01% of time. Instant classification success.

MCP TOOLS: 1009.8ms

99.9% of time. The primary optimization target.

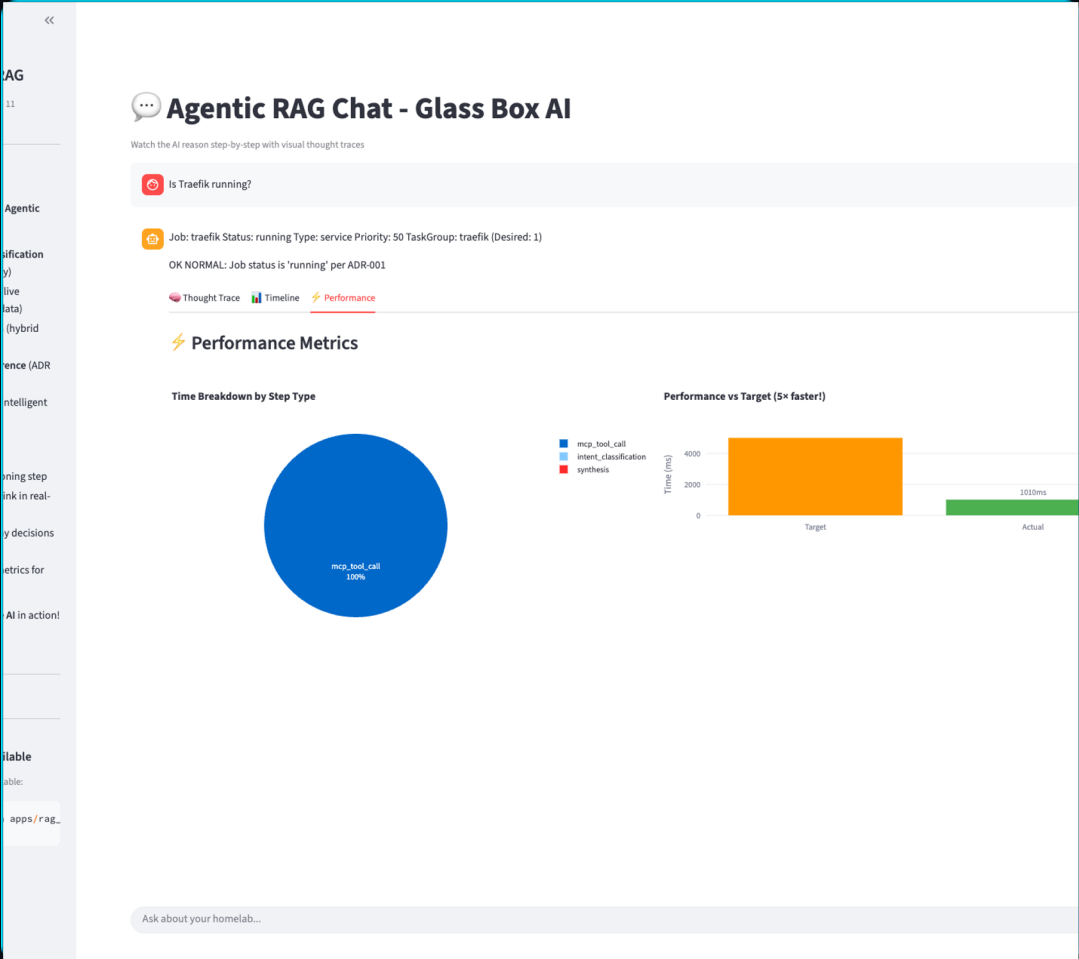
SYNTHESIS: 0.0ms

Instant final response generation.



Plotly visualization with real-time performance data

GLASS BOX AI: PERFORMANCE DASHBOARD ⚡



TIME BREAKDOWN

Pie chart proves MCP tool calls dominate the execution cycle.

TARGET VS ACTUAL

Goal: 5000ms
Actual: 1010ms
Result: 5× FASTER! 🏆

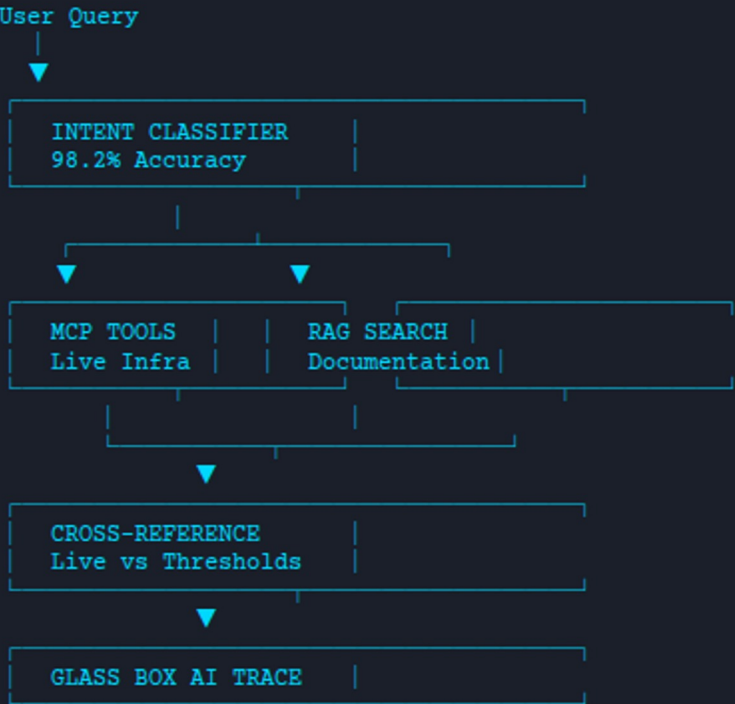
ENGINEERING EXCELLENCE

Proved where the bottleneck is via data-driven visualization.

Session statistics and intent distribution analytics

TECHNICAL ARCHITECTURE

SYSTEM FLOW



WORKFLOW FORMULA

Workflow = Visualize (Synthesize (Analyze (Execute (Classify (Q))))))

KEY METRICS

Intent Accuracy: 98.2%
Reasoning Success: 100%
Test Coverage: 190 Tests

KEY ENGINEERING PRINCIPLES

1. FOUNDATION

MCP tools before reasoning.
Reliable data enables smart workflows.

2. SECURITY

Safety isn't retrofitted. Built-in rate limits and read-only access.

3. INTENT IS KEY

Get routing right, and downstream success follows naturally.

4. SIMPLE WINS

Pattern matching vs ML. <1ms response. Complexity ≠ Better.

5. TRANSPARENCY

Visualization reveals real targets.
Optimized APIs, not AI.

6. MVP ITERATION

3 weeks, 3 distinct deliverables.
Low-risk progression.

Engineering is Methodology + Measurement +
Transparency

THE TRANSFORMATION

BEFORE PHASE 9

- ✓ Documentation Only
- ✓ 90% Retrieval Acc
- × No Live Data
- × Black Box Logic

ROLE: PASSIVE LIBRARY

AFTER PHASE 9

- ✓ Live Infrastructure
- ✓ 98.7% Agentic Acc
- ✓ Intent & Reasoning
- ✓ Visual Transparency

ROLE: INTELLIGENT CO-PILOT

98.7%

Accuracy

21-121ms

Response Time

190

Tests Passing

Time investment: 3 weeks (11 sessions, ~9 hours)

THE IMPACT

**3
WEEKS**

98.7%

ACCURACY

**41-238×
FASTER**

Systematic engineering transformed a static documentation system into an intelligent infrastructure co-pilot with complete visual transparency. The innovation: Not just what the AI can do, but letting users see HOW it does it.

METHODOLOGY

Foundation → Intelligence →
Transparency

INNOVATION

Glass Box AI visual reasoning
dashboard

NEXT

Phase 10: Continuing the evolution

Phase 9 Complete | Grade: A+ | 98.7% Agentic
Accuracy